

PURVAJ GHUDE

FULL-STACK ENGINEERING · APPLIED ML · CREATIVE TOOLS

Information Technology Undergraduate & Software Builder

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PROFILE

Information Technology undergraduate building measurable software across computer graphics, applied machine learning, AI systems, and mobile products. I turn ambitious ideas into documented, tested tools that expose how they work, where they perform well, and where they do not.

21 public GitHub repositories

123 contributions in the last year

99 automated Pixie tests

FLAGSHIP PROJECTS

TEJAS

Python · Machine learning · Space science · 2026

- Co-developed an Aditya-L1 solar-flare detection and forecasting pipeline. Processed 15,459 events with 97% GOES class agreement and 0.847 ROC-AUC using a leakage-free TCN + LightGBM ensemble.

PROJECT PROMETHEUS

Python · FastAPI · WebSockets · AI · 2026

- Built a persistent digital civilization of eight autonomous citizens with decisions, conversations, memories, relationships, and nightly reflection, plus a live Observer dashboard.

MESH

Flutter · Dart · Supabase · 2026

- Created a cross-platform collaboration app matching builders by complementary skills, with profiles, matching, chat, community feed, authentication, realtime flows, and migrations.

HERMES + LEDGER

Python · Docker · Data products · 2026

- Built a self-hosted opportunity engine discovering from seven sources every six hours, then added Telegram alerts, price tracking, and a privacy-first finance tracker.

PIXIE WIXIE

TypeScript · Computer graphics · 2026

- Built an offline raster-to-vector studio and CLI with inspectable geometry; reduced one mind-map from 29,431 paths to 6 paths and 17.4 KB, backed by 99 automated tests.

GAME OF GROWTH

React · FastAPI · MongoDB · 2026

- Created a creative-agency experience with 12+ visual sections and a founder CRM; documented 17/17 passing API tests and end-to-end flows.

GESTURE-CONTROLLED PARTICLES

JavaScript · Computer vision · Creative coding · 2025

- Built a browser experiment mapping hand gestures to real-time particle behavior through low-latency canvas interaction.

ENGINEERING HIGHLIGHTS

- Published TEJAS models and held-out predictions so forecasting results can be independently verified.
- Designed Pixie Wixie around real intermediate algorithm state, making every transformation inspectable.
- Built across browser, CLI, mobile, backend, database, realtime, AI, and machine-learning layers.
- Documented setup, architecture, testing, limitations, and contribution workflows for reproducibility.

TECHNICAL RANGE

Languages: TypeScript, Python, Dart, JavaScript, HTML/CSS, SQL, Java

Frameworks & platforms: React, Vite, Flutter, FastAPI, Supabase, Firebase, MongoDB, Node.js, Streamlit

Engineering: Algorithms, computer graphics, machine learning, AI product design, API design, WebSockets, data pipelines, automated testing, benchmarking, Docker, Git/GitHub

EDUCATION

B.E. Information Technology

University of Mumbai · Mahatma Gandhi Missions College of Engineering and Technology · 2025—2029

HACKATHONS

Bharatiya Antariksh Hackathon · Aditya-L1 solar-flare challenge · 2026

RedRob Ideathon · Mesh collaboration app · 2026

INTERESTS & HOBBIES

Computer graphics · Algorithm visualization · Space science · AI simulations · Technical writing · Learning in public